

Qatar-1b

R-2689

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_110620 · created 2026-08-31T01:17:06+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2455732.83526 +/- 0.00042 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1593 +/- 0.0092
Transit depth (Rp/Rs)^2	2.54 +/- 0.29 [%]
Orbital Inclination (inc)	86.06 +/- 1.16
Airmass coefficient 1 (a1)	1.311 +/- 0.015
Airmass coefficient 2 (a2)	-0.212 +/- 0.019
Scatter in the residuals of the lightcurve fit is	1.11 %
Best Comparison Star	None
Optimal Aperture	3.07
Optimal Annulus	10.47
Transit Duration (day)	0.0778 +/- 0.0037

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1593 ± 0.0092 vs 0.1463 (deviation 1.07σ)
Duration fit vs published	0.0778 d vs 0.0692 d (deviation 1.77σ)
Mid-time O–C	1.1 min
Depth SNR	13.1
Residual scatter	1.11 %

Light curve

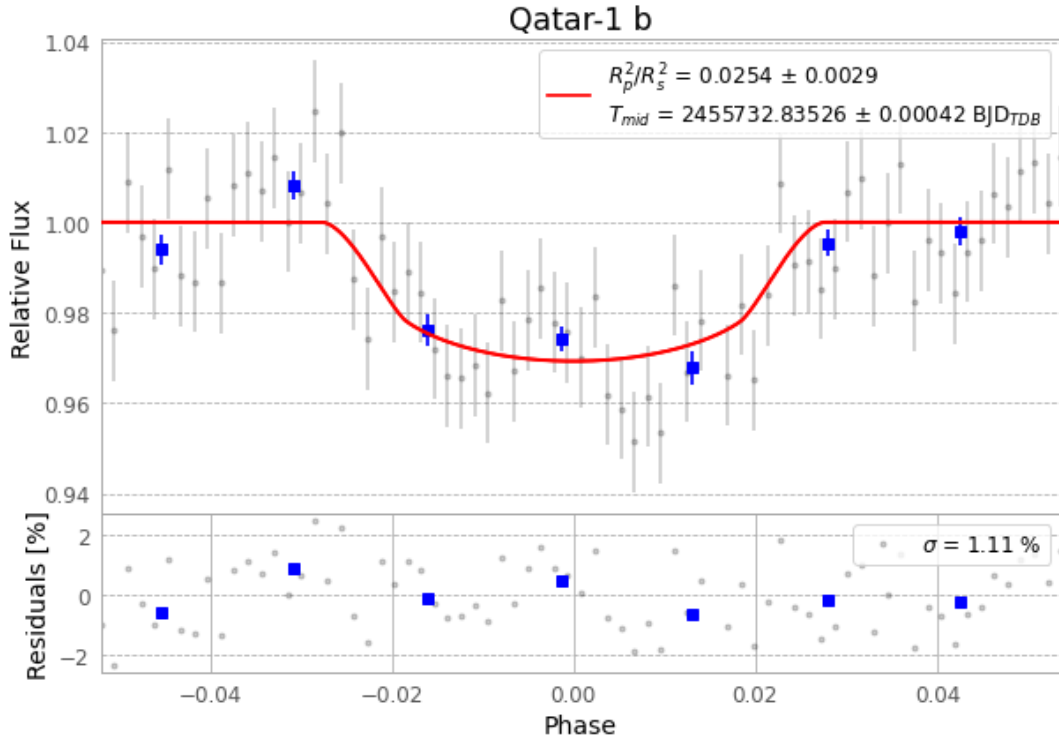


Figure 1 — FinalLightCurve_Qatar-1 b_2011-06-19.png (SHA-256 27e5c4b586489f43...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [415, 255], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 de8c946523f24081...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

73 FITS frames (48 MB), Qatar-1110620061513.FITS → Qatar-1110620095113.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-110620.log (c039dc55d31c...), FinalLightCurve_Qatar-1 b_2011-06-19.png (27e5c4b58648...), FinalLightCurve_Qatar-1 b_2011-06-19.csv (edae528c526c...)

Compute resources

Wall time 153.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-31 01:17Z.